

Digital Image Processing (CS-323)

Lecture-9

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- Topics to Learn

1. Spatial Filtering

- Mechanics of Spatial Filtering
- Spatial Correlation and Convolution
- Smoothing Linear Filters
- Order-Statics (Nonlinear) Filters

Spatial Filtering: Mechanics of Spatial Filtering

- The name filter is borrowed from frequency domain processing where “**filtering**” refers to passing, modifying, or rejecting specified frequency components of an image.
- For example, a **filter that passes low frequencies is called a lowpass filter.**
- The net effect produced by a lowpass filter is to **smooth** an image by blurring it.
- We can accomplish similar smoothing directly on the image itself by using spatial filters.

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Spatial Filtering: Mechanics of Spatial Filtering

- The **Spatial filtering** modifies an image by replacing the value of each pixel by a function of the values of the pixel and its neighbors.
- If the **operation performed on the image pixels is linear**, then the filter is called a **linear spatial filter**.
- Otherwise, the filter is a **nonlinear spatial filter**.

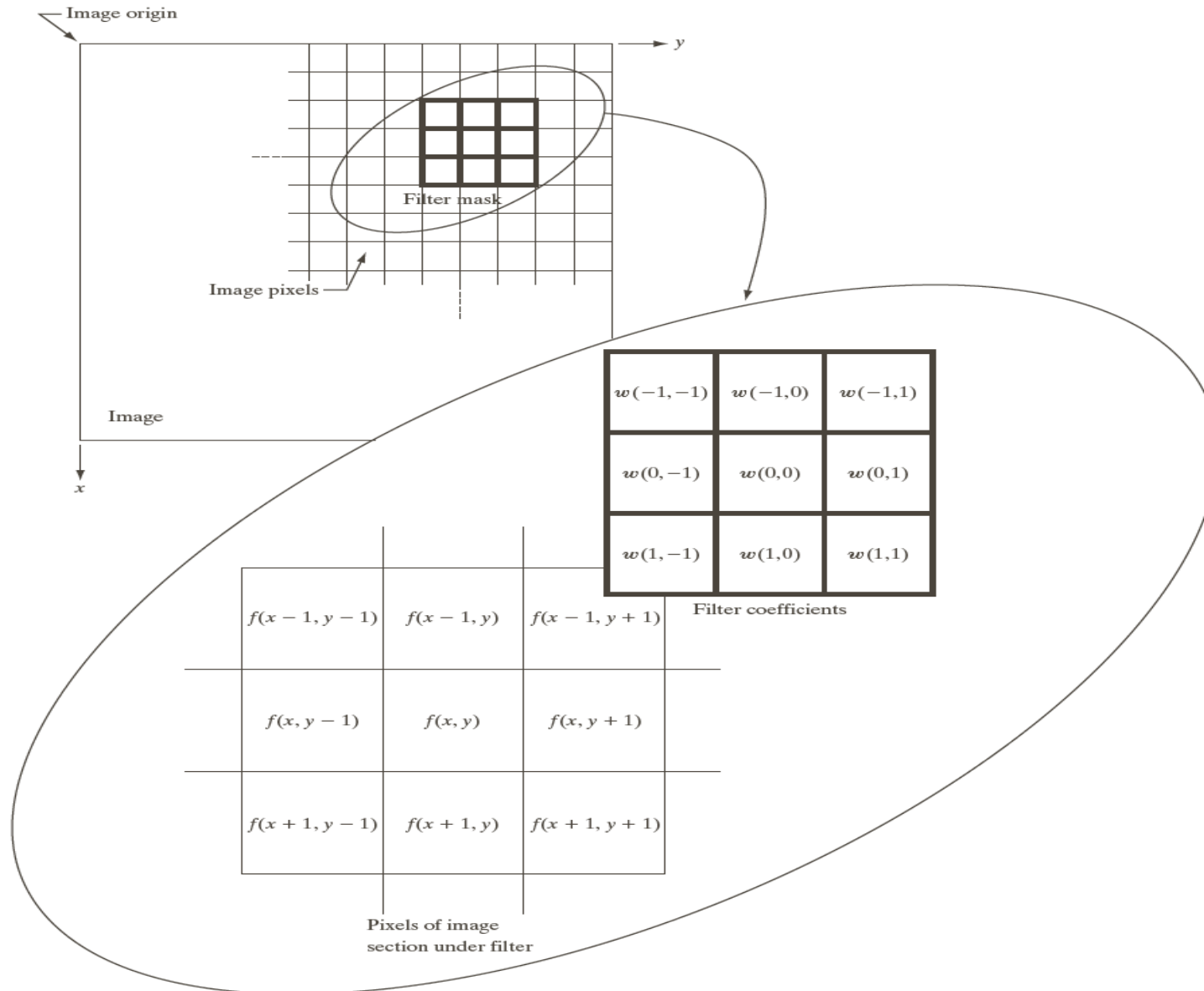
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Spatial Filtering: Mechanics of Spatial Filtering

- A linear spatial filter performs a sum-of-products operation between an image f and a filter kernel, w .
- The kernel is an array whose size defines the neighborhood of operation, and whose coefficients determine the nature of the filter.
- Other terms used to refer to a spatial filter kernel are mask, template, and window.

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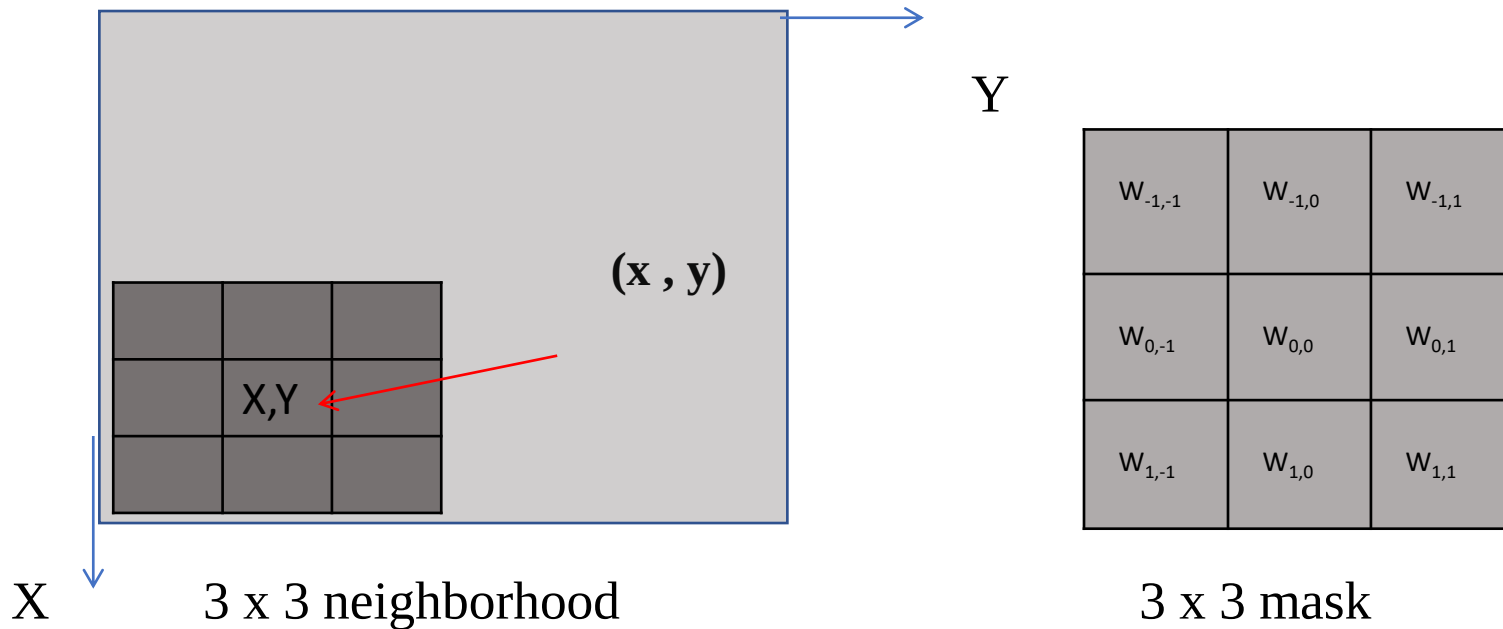
Spatial Filtering: Mechanics of Spatial Filtering



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Spatial Filtering: Mechanics of Spatial Filtering

Mask Processing



$$g(x, y) = \sum_{i=-1}^1 \sum_{j=-1}^1 w_{i,j} f(x + i, y + j)$$

- $G(x, y) = w(-1, -1)f(x-1, y-1) + w(-1, 0)f(x-1, y) + \dots$
 $\dots + w(0, 0)f(x, y) + \dots + w(1, 1)f(x+1, y+1)$

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Spatial Filtering: Mechanics of Spatial Filtering

- The center coefficient of the kernel, $w(0, 0)$ aligns with the pixel at location (x, y) .
- For a kernel of size $m \times n$, we assume that $m = 2a + 1$ and $n = 2b + 1$, where a and b are nonnegative integers.
- This means that our focus is on **kernels of odd size** in both coordinate directions.

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Spatial Filtering: Spatial Correlation and Convolution

- Correlation consists of moving the center of a kernel over an image, and computing the sum of products at each location.
- The mechanics of *spatial convolution* are the same, except that the correlation kernel is rotated by 180° .
- Thus, when the values of a kernel are symmetric about its center, correlation and convolution yield the same result.

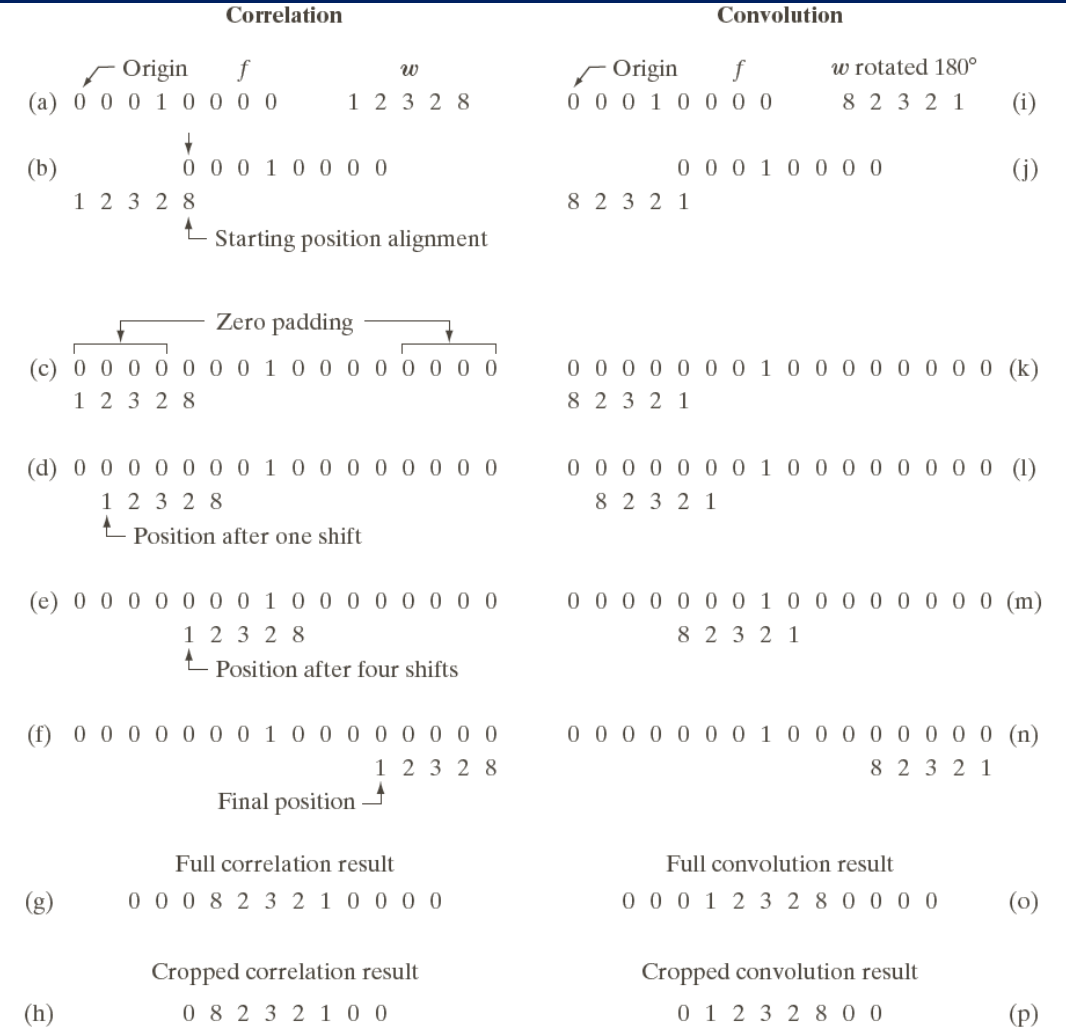


FIGURE 3.29 Illustration of 1-D correlation and convolution of a filter with a discrete unit impulse. Note that correlation and convolution are functions of *displacement*.

Spatial Filtering: Spatial Correlation and Convolution

- There are two important points to note from the preceding analysis.
- First, correlation is a **function of displacement** of the filter kernel relative to the image.
- The second thing to notice is that correlating a kernel w with a function that contains all 0's and a single 1 **yields a copy of w** , but **rotated by 180°** .
- A function that contains a single 1 with the rest being 0's is called a **discrete unit impulse**.
- **Convolution function** with an impulse yields a copy of the function at the location of the impulse.

Spatial Filtering: Spatial Correlation and Convolution

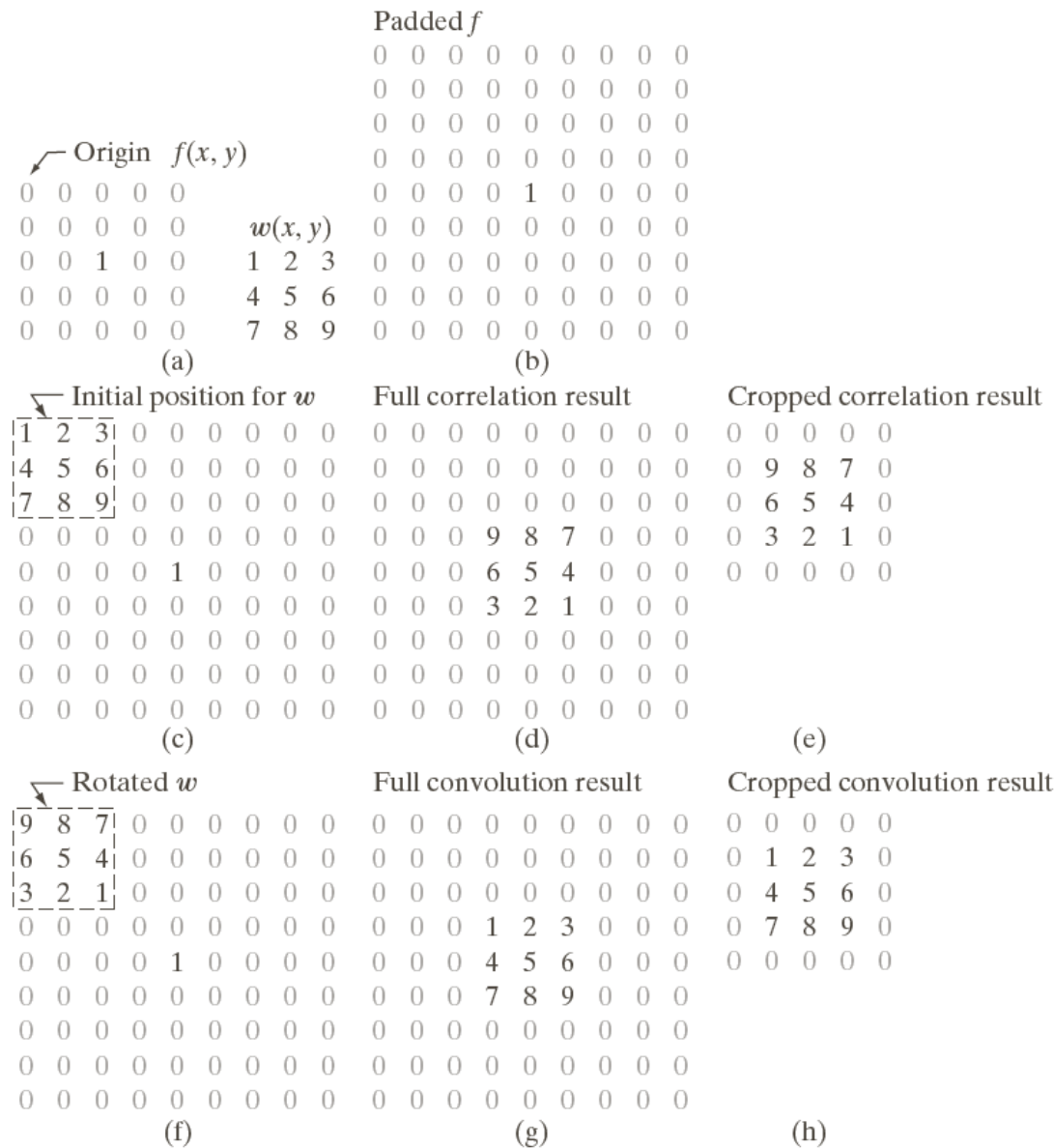


FIGURE 3.30
Correlation
(middle row) and
convolution (last
row) of a 2-D
filter with a 2-D
discrete, unit
impulse. The 0s
are shown in gray
to simplify visual
analysis.

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Spatial Filtering: Smoothing Linear Filters

- The term linear spatial filtering, we mean convolving a kernel with an image.
- In smooting (averaging) filter, we repace the central pixel by neighbourhood average.
- Help is reduce random sharp transitions. However, edges are also sharp tansitions.

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

$$\frac{1}{16} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.

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Spatial Filtering: Order-Statics (Nonlinear) Filters

- **Order-statistic filters are nonlinear** spatial filters whose response is based on ordering (ranking) the pixels contained in the region encompassed by the filter.
- Smoothing is achieved by replacing the value of the center pixel with the value determined by the ranking result.
- The best-known filter in this category is the **median filter**, which, as its name implies, replaces the value of the center pixel by the median of the intensity values in the neighborhood of that pixel

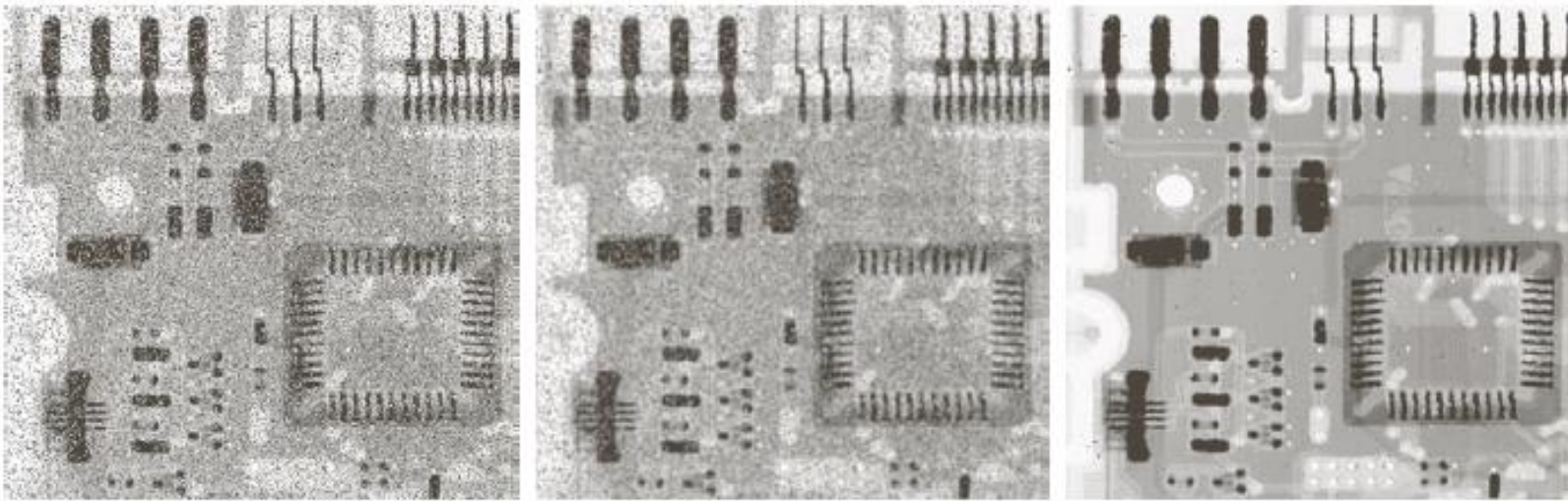
Spatial Filtering: Order-Statics (Nonlinear) Filters

- Median filters provide excellent **noise reduction capabilities** for certain types of random noise, with considerably **less blurring**.
- Median filters are particularly effective in the presence of **impulse noise** (sometimes called salt-and-pepper noise).
- The median represents the **50th percentile** of a ranked set of numbers, but ranking lends itself to many other possibilities.
- Thus, the principal function of median filters is to force points to be more like their neighbors.

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Spatial Filtering: Order-Statics (Nonlinear) Filters

- The **100th percentile** results in the so-called **max filter**, which is useful for finding the brightest points in an image or for eroding dark areas adjacent to light regions.
- The **0th percentile** filter is the **min filter**, used for the opposite purpose.



a b c

FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

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